

n-selection / depth-selection strategies

Question: do STE / REINFORCE compete with SoftGating / Gumbel on dynamic-k (ProbReg) and dynamic-depth (FlexNN)?

Expectation:

- SoftGating: best baseline (smooth weighted average; always in scope).
- Gumbel-Softmax: equivalent to SoftGating with extra sampling noise; may be worse on small data.
- STE: comparable to SoftGating in theory (straight-through estimator).
- REINFORCE: high variance; needs a baseline subtraction to be competitive.

A separate ablation (no-regularisation) checks whether the dynamic-k predictor actually adapts to the data when not pinned by the ELBO prior.

ProbReg dyn-k (ELBO): best cell per (dataset, selection)

dataset	strategy	selection	regularization	mse	nll	crps
b1_gravitational	separate_heads	reinforce	elbo	0.293	0.601	0.192
bimodal	separate_heads	reinforce	elbo	2.328	1.847	0.915
bimodal	separate_heads	soft_gating	elbo	2.358	1.848	0.921
bimodal	separate_heads	ste	elbo	2.357	1.853	0.915
heteroscedastic	separate_heads	reinforce	elbo	2.488	1.859	0.843
heteroscedastic	separate_heads	soft_gating	elbo	1.623	1.378	0.639
heteroscedastic	separate_heads	ste	elbo	2.482	1.852	0.840

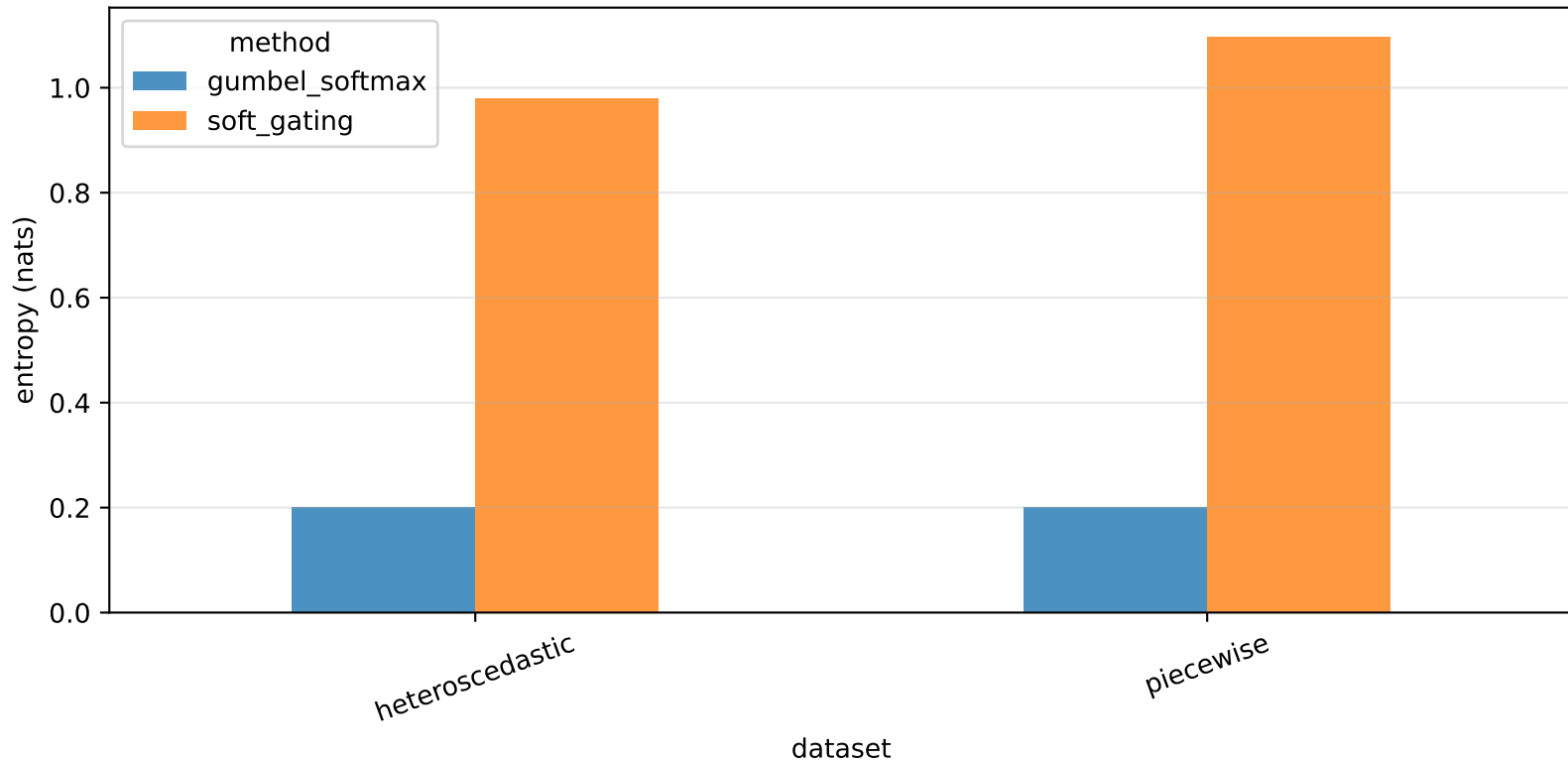
FlexNN dyn-depth: best cell per (dataset, layer_method)

dataset	weights	layer_method	depth_reg	mse	mean_depth
b3_two_phase	shared	gumbel_softmax	depth_penalty	0.085	1.476
b3_two_phase	shared	soft_gating	cost_aware_elbo	0.082	1.772
b3_two_phase	shared	ste	elbo	0.085	2.628
piecewise	shared	gumbel_softmax	elbo	0.265	1.373
piecewise	shared	reinforce	none	0.260	--
piecewise	shared	soft_gating	elbo	0.261	2.006
piecewise	shared	ste	cost_aware_elbo	0.273	2.492

Gumbel vs SoftGating + ELBO retest (post-fix)

dataset	model	method	reg	mse	nll	mean_dept	mean_k	entropy
piecewise	FlexNN	soft_gating	none	0.255	--	1.043	--	0.041
piecewise	FlexNN	soft_gating	elbo	0.264	--	2.088	--	1.390
piecewise	FlexNN	Gumbel_softmax	none	0.270	--	1.002	--	0.007
piecewise	FlexNN	Gumbel_softmax	elbo	0.265	--	1.373	--	0.322
piecewise	ProbReg	soft_gating	none	0.261	0.486	--	3.863	1.182
piecewise	ProbReg	soft_gating	elbo	0.267	0.507	--	2.442	1.777
piecewise	ProbReg	Gumbel_softmax	none	0.281	0.560	--	5.425	0.005
piecewise	ProbReg	Gumbel_softmax	elbo	0.270	0.475	--	2.000	0.471
heterosced	FlexNN	soft_gating	none	1.696	--	1.726	--	0.148
heterosced	FlexNN	soft_gating	elbo	1.620	--	2.072	--	1.356
heterosced	FlexNN	Gumbel_softmax	none	1.609	--	1.803	--	0.037
heterosced	FlexNN	Gumbel_softmax	elbo	1.818	--	1.188	--	0.167
heterosced	ProbReg	soft_gating	none	1.633	1.405	2102744	48.000	0.712
heterosced	ProbReg	soft_gating	elbo	1.623	1.378	--	2.000	1.702
heterosced	ProbReg	Gumbel_softmax	none	1.757	1.446	27733	328.000	0.044
heterosced	ProbReg	Gumbel_softmax	elbo	2.416	1.739	--	2.000	0.558

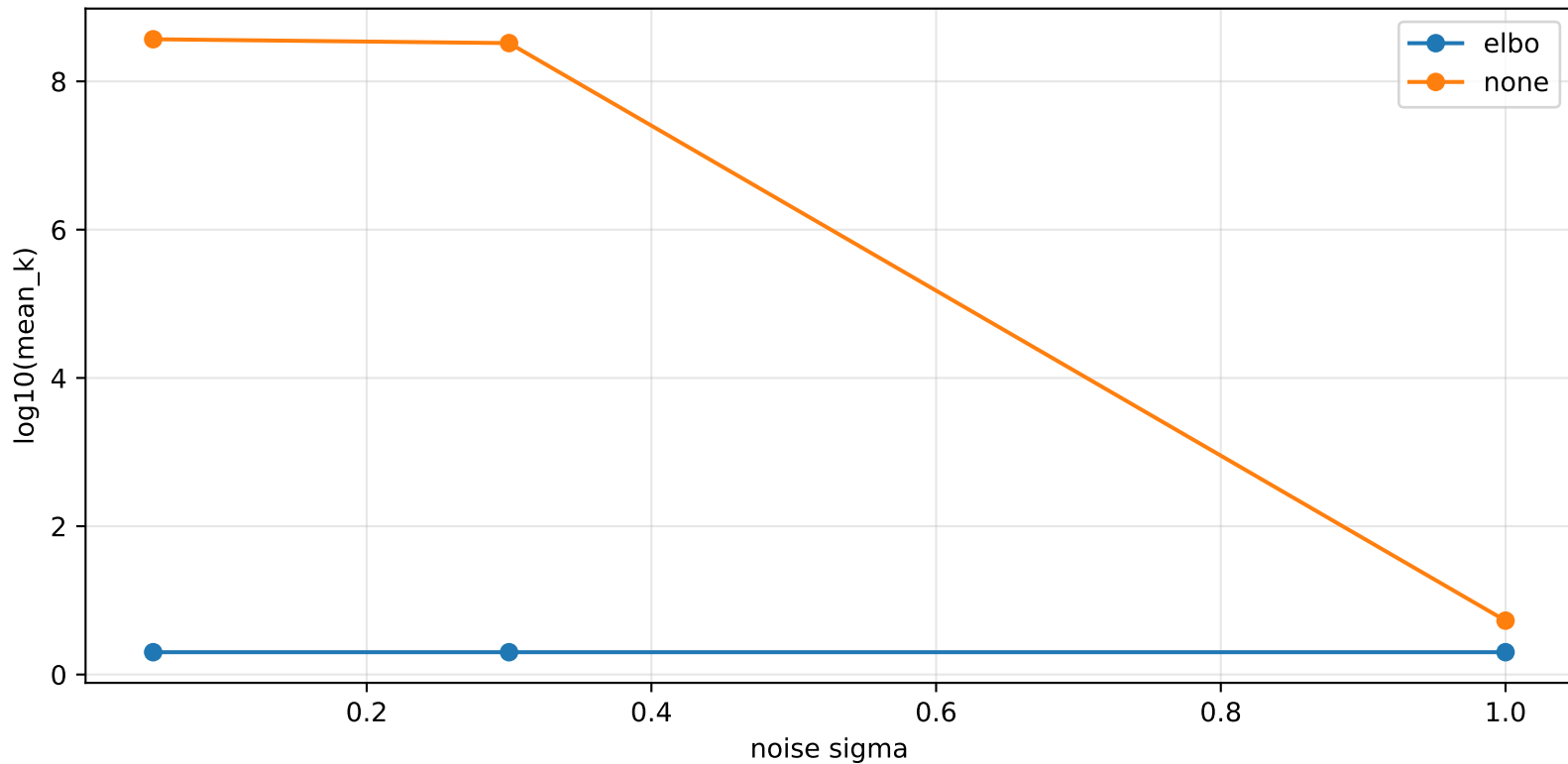
Selection-probability entropy (higher = more input-dependent)



ProbReg dyn-k adaptation (noise bench): mean_k vs sigma

sigma	max_k	reg	mean_k	mse
0.050	10	elbo	2.000	0.012
0.050	10	none	369844416.000	0.006
0.300	10	elbo	2.000	0.134
0.300	10	none	328087776.000	0.120
1.000	10	elbo	2.000	1.099
1.000	10	none	5.328	1.111

Dynamic-k ProbReg: mean selected k vs noise



Verdict — n-selection

Verdict: PARTIAL: With ELBO regularisation the dynamic-k predictor collapses to $k=2$ n